1. Main Java Program (Single File)

Save as: LibraryManagementSystem.java

```
import java.util.*;

class LibraryManagementSystem {

    static final int ALLOWED_DAYS = 14;
    static final int FINE_PER_DAY = 5;

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        // User details
        System.out.print("Enter User Name: ");
        String name = sc.nextLine();

        System.out.print("Enter User ID: ");
        int id = sc.nextInt();

        System.out.print("Enter number of books issued: ");
        int books = sc.nextInt();

        sc.nextLine(); // consume newline

        List<String> bookList = new ArrayList<>();

        for (int i = 0; i < books; i++) {
            System.out.print("Enter book name " + (i + 1) + ": ");
            bookList.add(sc.nextLine());
        }

        System.out.print("Enter number of days books were borrowed: ");
        int days = sc.nextInt();

        // Fine calculation
        int fine = 0;
        if (days > ALLOWED_DAYS) {
            fine = (days - ALLOWED_DAYS) * FINE_PER_DAY;
        }

        // Output
        System.out.println("\n===== LIBRARY DETAILS =====");
        System.out.println("Name: " + name);
        System.out.println("ID: " + id);
        System.out.println("Books Issued: " + books);
        System.out.println("Book List: " + bookList);
        System.out.println("Days Borrowed: " + days);
```

```
        System.out.println("Total Fine: Rs. " + fine);
    }
}
```

✅ 2. Test

Test File (JUnit)

Save as: TestLibrary.java

```
import org.junit.Test;
import static org.junit.Assert.*;

public class TestLibrary {

    @Test
    public void testFineCalculation() {
        int allowedDays = 14;
        int finePerDay = 5;

        int days = 20;
        int expectedFine = (days - allowedDays) * finePerDay;

        int actualFine = (days > allowedDays) ?
                    (days - allowedDays) * finePerDay : 0;

        assertEquals(expectedFine, actualFine);
    }

    @Test
    public void testNoFine() {
        int days = 10;
        int allowedDays = 14;

        int fine = (days > allowedDays) ? (days - allowedDays) * 5 : 0;

        assertEquals(0, fine);
    }
}
```

✅ 3. pom.xml (Maven Configuration)

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
        xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
        xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
        http://maven.apache.org/xsd/maven-4.0.0.xsd">
```

```
    <modelVersion>4.0.0</modelVersion>

    <groupId>com.library</groupId>
    <artifactId>library-management</artifactId>
    <version>1.0-SNAPSHOT</version>

    <dependencies>
      <!-- JUnit for testing -->
      <dependency>
        <groupId>junit</groupId>
        <artifactId>junit</artifactId>
        <version>4.13.2</version>
        <scope>test</scope>
      </dependency>
    </dependencies>

    <build>
      <plugins>
        <!-- Compiler Plugin -->
        <plugin>
          <artifactId>maven-compiler-plugin</artifactId>
          <version>3.8.1</version>
          <configuration>
            <source>1.8</source>
            <target>1.8</target>
          </configuration>
        </plugin>
      </plugins>
    </build>

</project>
```

✅ 4. Jenkins Pipeline Script

Save as: Jenkinsfile

```
pipeline {
    agent any


    tools {
      jdk 'JDK17'         // Name configured in Jenkins Global Tool Configuration
      maven 'Maven3'      // Name configured in Jenkins
    }
```

```
stages {

    stage('Clone Repository') {
        steps {
            git 'https://github.com/your-username/library-management.git'
        }
    }

    stage('Build') {
        steps {
            bat 'mvn clean compile'
        }
    }

    stage('Test') {
        steps {
            bat 'mvn test'
        }
    }

    stage('Package') {
        steps {
            bat 'mvn package'
        }
    }
}

post {
    success {
        echo 'Build Successful!'
    }
```

```
        failure {
            echo 'Build Failed!'
        }
    }
}
```